Status	Finished
Started	Monday, 26 October 2026, 4:36 PM
Completed	Monday, 26 October 2026, 4:41 PM
Duration	4 mins 42 secs
Grade	**14.00** out of 15.00 (**93.33**%)

Question **1**

Complete

Mark 0.00 out of 1.00

Which algorithm is efficient for analyzing the frequency distribution of items in a list?

- a. Quick Sort
- b. Bubble Sort
- c. Merge Sort
- d. Linear Search

The correct answer is: Merge Sort

Question **2**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm involves comparing elements and swapping adjacent items that are out of order?

- a. Merge Sort
- b. Bubble Sort
- c. Binary Search
- d. Linear Search

The correct answer is: Bubble Sort

Question **3**

Complete

Mark 1.00 out of 1.00

Why is it advantageous to sort data before performing duplicate analysis?

- a. It makes the analysis slower
- b. It complicates the analysis process
- c. It has no effect on the analysis process
- d. It allows for quicker identification of duplicates

The correct answer is: It allows for quicker identification of duplicates

Question **4**

Complete

Mark 1.00 out of 1.00

Which of the following best describes the term "sorting" in computer science?

- a. Arranging data in a specific order
- b. Merging two datasets
- c. Removing duplicates from a list
- d. Finding a specific element in a list

The correct answer is: Arranging data in a specific order

Question **5**

Complete

Mark 1.00 out of 1.00

Why is sorting important for selection operations?

- a. It slows down the process
- b. It makes it easier to select items based on their relationship to the rest of the items
- c. It makes the data unsorted
- d. It complicates the selection of items

The correct answer is: It makes it easier to select items based on their relationship to the rest of the items

Question **6**

Complete

Mark 1.00 out of 1.00

Algorithm design technique used in merge sort algorithm is

- a. Divide and conquer
- b. Backtracking
- c. Greedy method
- d. Dynamic programming

The correct answer is:
Divide and conquer

Question **7**

Complete

Mark 1.00 out of 1.00

What type of problems can sorting help solve efficiently?

a. Duplicates
b. All of the above
c. Selection
d. Searching

The correct answer is: All of the above

Question **8**

Complete

Mark 1.00 out of 1.00

What is the primary benefit of using sorting algorithms in programming?

a. Decreases the efficiency of algorithms
b. Makes code execution slower
c. Makes data harder to manage
d. Provides a basis for other algorithms to work efficiently

The correct answer is: Provides a basis for other algorithms to work efficiently

Question **9**

Complete

Mark 1.00 out of 1.00

In Merge Sort, what happens after dividing the input into smaller parts?

a. Each part is searched for a specific element
b. Each part is sorted independently
c. The parts are ignored
d. The parts are merged without sorting

The correct answer is: Each part is sorted independently

Question **10**

Complete

Mark 1.00 out of 1.00

In the context of sorting, what does the divide-and-conquer approach involve?

- a. Dividing the input into parts, solving each part, and combining the solutions
- b. Rearranging data without sorting
- c. Sorting data in a single pass
- d. Sorting data sequentially

The correct answer is: Dividing the input into parts, solving each part, and combining the solutions

Question **11**

Complete

Mark 1.00 out of 1.00

Which of the following is not an in-place sorting algorithm?

- a. Heap sort
- b. Merge sort
- c. Quick sort
- d. Selection sort

The correct answer is:
Merge sort

Question **12**

Complete

Mark 1.00 out of 1.00

_____ explain how an algorithm will perform when the input grows larger.

- a. Searching
- b. Sorting
- c. Complexity
- d. Merging

The correct answer is:
Complexity

Question **13**

Complete

Mark 1.00 out of 1.00

The process of placing or rearranging a collection of elements into a particular order is known as

a. Rearranging

b. Sorting

c. Searching

d. Merging

The correct answer is: Sorting

Question **14**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm is based on repeatedly dividing the list into halves?

a. Merge Sort

b. Insertion Sort

c. Quick Sort

d. Bubble Sort

The correct answer is: Merge Sort

Question **15**

Complete

Mark 1.00 out of 1.00

Which sorting algorithm would be preferred for its divide-and-conquer approach?

a. Bubble Sort

b. Linear Search

c. Binary Search

d. Merge Sort

The correct answer is: Merge Sort